17th of October, 2026

The Nordic Testicular Cancer Gap Has More Than Halved

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Project

Descriptive epidemiology, built in R
Structured by the CRISP-DM phases

Data

NORDCAN 2.0, data version 9.6 (06.2026) Males,
cancer of the testis, five Nordic countries

Two Numbers, Thirty Years Apart

The highest country went from 3.8 times the lowest country's rate to 1.8 times.

3.82

1990-94: Denmark's rate against Finland's (10.0 and 2.6 per 100,000)

1.83

2020-24: Denmark's rate against Iceland's (10.6 and 5.8 per 100,000)

52%

Fall in the high-to-low ratio between the two periods



Same five countries, same registry, same method, thirty years apart

The two countries at the extremes differ between the periods, because the extremes change over time.

What Stays the Same

5 Nordic countries: Denmark, Finland, Iceland, Norway, Sweden

1 Registry: NORDCAN 2.0, data version 9.6

1 Method: age-standardised rates, ESP2013

Business Understanding

The question is how big the gap is and whether it narrowed, a gap that was noticed but never measured.

Why This Question

Research Paper Problem

A published Nordic survival study of male cancers (Tichanek et al., 2023, Cancers) notes in its Discussion that national differences in testicular cancer incidence are not well understood, and that Danish rates are known to be some of the highest in the world.

Project Scope

That study measured survival in four countries, without Iceland. It did not investigate the incidence gap.

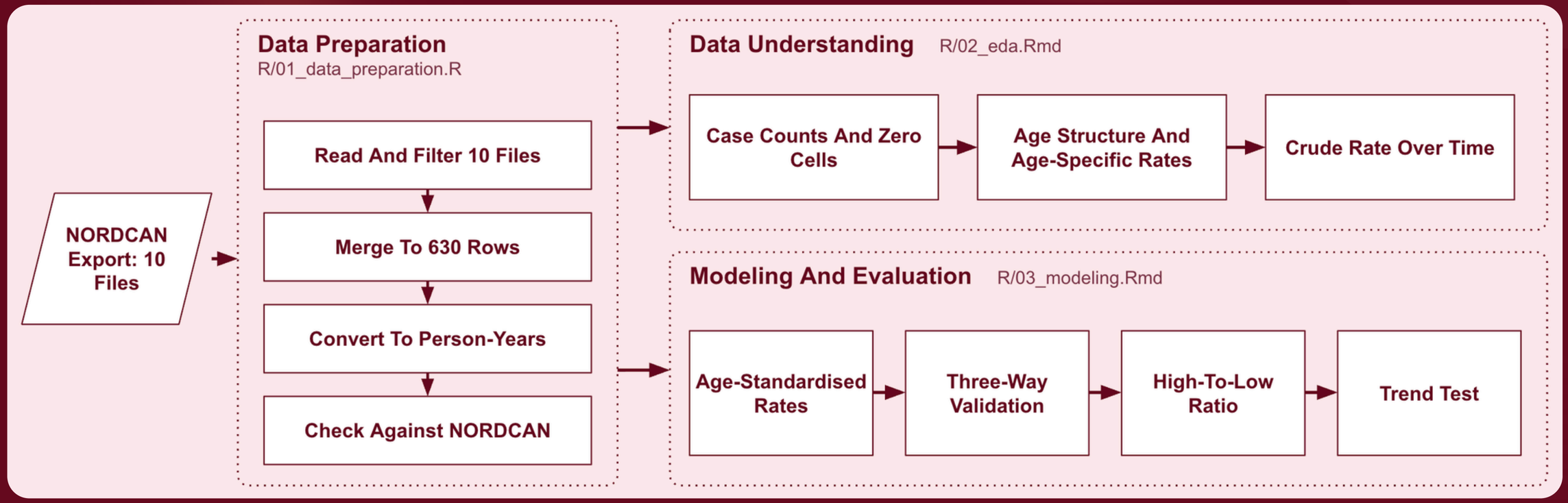
Research Question

How does age-standardised testicular cancer incidence compare across the five core Nordic countries, and has the gap between the highest and lowest country narrowed or persisted over 1990 to 2024?

What Counts As An Answer

- One comparable rate for every country and period
- A high-to-low ratio per period, with its interval
- A formal test of whether the ratio is falling

Workflow


NORDCAN Export: 10 Files
Data Preparation
R/01_data_preparation.R
Read And Filter 10 Files
Merge To 630 Rows
Convert To Person-Years
Check Against NORDCAN
Data Understanding
R/02_eda.Rmd
Case Counts And Zero Cells
Age Structure And Age-Specific Rates
Crude Rate Over Time
Modeling And Evaluation
R/03_modeling.Rmd
Age-Standardised Rates
Three-Way Validation
High-To-Low Ratio
Trend Test

Data Understanding

Four countries bring hundreds to thousands of cases per period. Iceland brings 38 to 73, so its figures are the least precise and stay visible throughout.

34,568 cases in total

Males, cancer of the testis, 1990 to 2024

360 of those cases are Iceland's

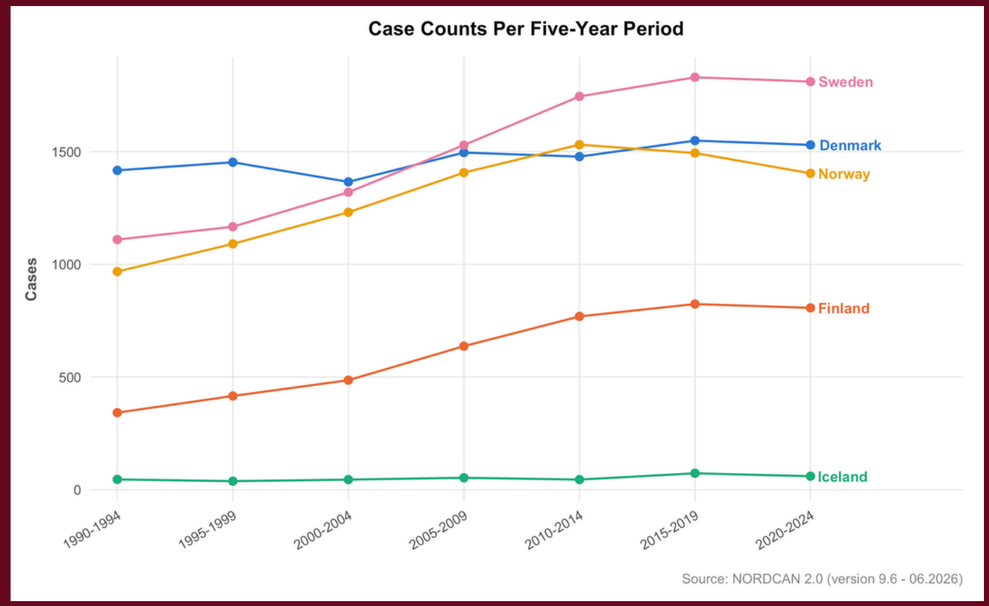
38 to 73 cases in total

against 342 to 1,830 in each of the other four

74 of 630 cells

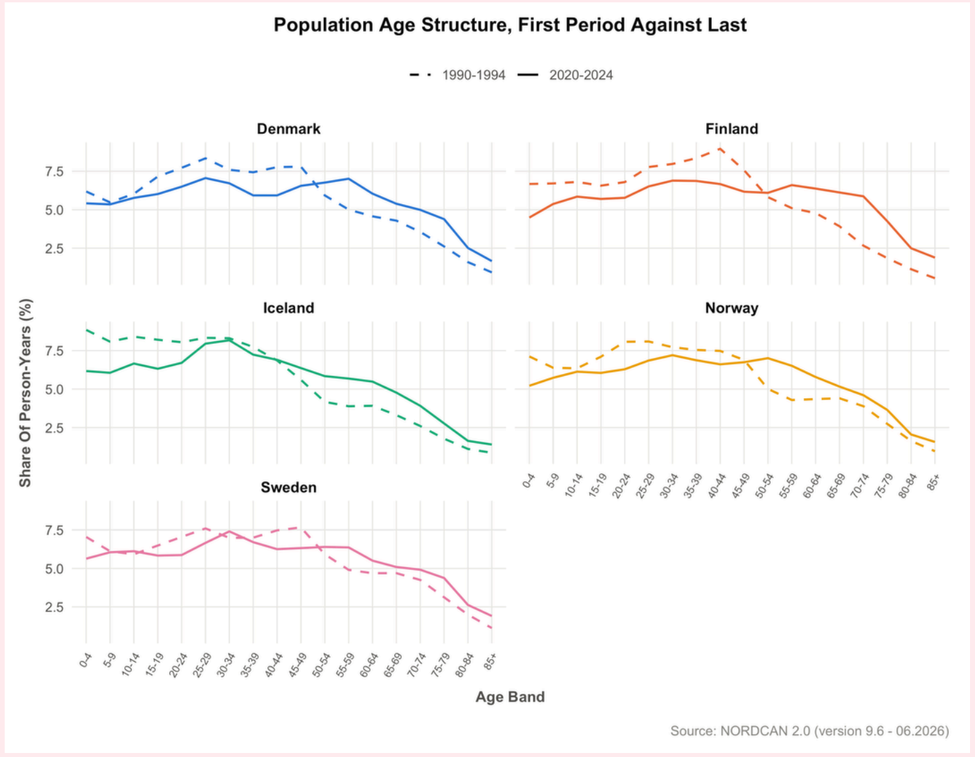
hold no cases, 48 of them Iceland's

Cases Per Five-Year Period, By Country



The Populations Are Not The Same Age

Share Of Person-Years By Age, First Period Against Last



Share Of Person-Years

Country	Under 35, 1990-94	Under 35, 2020-24	65 And Over, 1990-94	65 And Over, 2020-24
Denmark	48.5%	42.8%	13.0%	19.0%
Finland	49.3%	40.6%	10.1%	20.6%
Iceland	58.2%	48.0%	9.7%	14.5%
Norway	50.8%	43.5%	13.6%	17.0%
Sweden	47.2%	43.6%	15.2%	18.9%

Peak At 25 To 34

In every country. Countries differ in how high it goes, not when.

Finland Ages Fastest

Its 65 and over share doubles, 10.1% to 20.6%.

Data Preparation for Analysis

The data is checked against NORDCAN's own published figures before any analysis starts.

Read And Filter

10 NORDCAN files, kept to the five study countries.

Merge

Case counts and populations joined into one table of 630 rows: 5 countries, 7 periods, 18 age bands.

Cross-Check

Case totals match NORDCAN's own in all 35 country-periods. All 630 age-specific rates reproduce NORDCAN's published rates once rounded.

Person-Years

NORDCAN gives an average annual population. Each period spans five years, so it is multiplied by 5.

Problem Encountered & Addressed:
Every Rate Comes Out 5 Times Too Large

Output

A 630-row analysis table and a 35-row answer key of NORDCAN's published figures. The raw data is not republished here. The repository holds the code and the results.

From Five Rates To One Gap Number

The highest country went from 3.8 times the lowest country's rate to 1.8 times, two numbers that are thirty years apart.

Input

Five countries' age-specific rates, across seven periods

Output

One gap number per period, and a test of its trend

Age-Standardised Rate

- Each country's rate is reweighted onto one shared reference population, ESP2013, the one NORDCAN uses.
- Its 19 age groups end at 90+ and the data ends at 85+, so the top two weights are merged. Intervals at 95%, with 99% as a stricter check.

High-To-Low Ratio

- In each period, the highest country's rate divided by the lowest.
- It is recomputed every period, because the extremes change: four different pairings appear across the seven periods.
- Each ratio has its own interval.

Trend Test

- A straight line through the log of the ratio against each period's midpoint year, with precise periods counting for more.
- It asks whether the ratio is falling instead of reading a slope off a chart.

Tools: R, dplyr, ggplot2, PHEindicatormethods, broom, renv, Git

The Arithmetic Holds Up Three Ways

Three routes land on the same number, and the outside check sits more than 300 times inside the tolerance.



0.00015
Largest difference from NORDCAN's published rates, against an allowed 0.05

Three Checks

< 1e-9

Hand calculation against phe_dsr()

Near Exact

Pass

< 1e-9

phe_dsr() against calculate_dsr()

Near Exact

Pass

0.00015

Computed rate against NORDCAN's published rate

Allowed 0.05

Pass

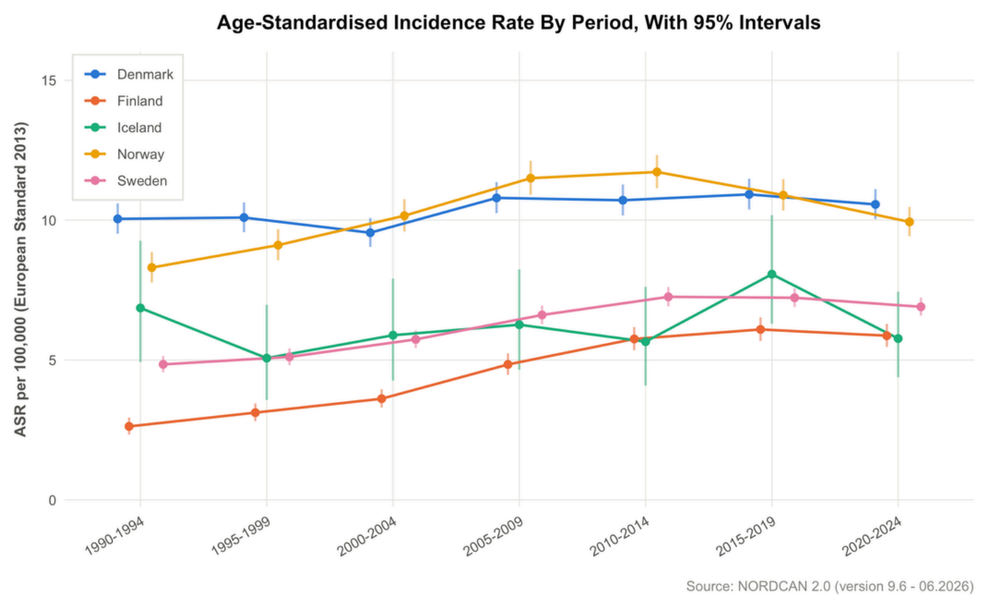


Insights

- Both tolerances were set before any computation ran.
- Denmark 1990-94 comes out at 10.04845 by hand, from both packages, and in NORDCAN's table.
- The check itself was tested: adding 0.06 to the computed rates fails it, adding 0.04 passes.
- Swapping two neighbouring standard-population weights slips past the first four guards and still lands inside the tolerance (0.0463 against 0.05). A check on the weights themselves now closes that gap.

The Bottom Rose While The Top Held

Standardised Rate Per 100,000, With 95% Intervals



First Period Against Last

Country	1990 - 1994	2020 - 2024	Change
Finland	2.63	5.87	+123%
Sweden	4.85	6.90	+42%
Norway	8.31	9.94	+20%
Denmark	10.05	10.56	+5%
Iceland	6.86	5.77	-16%

Largest Rise
Finland, +123%

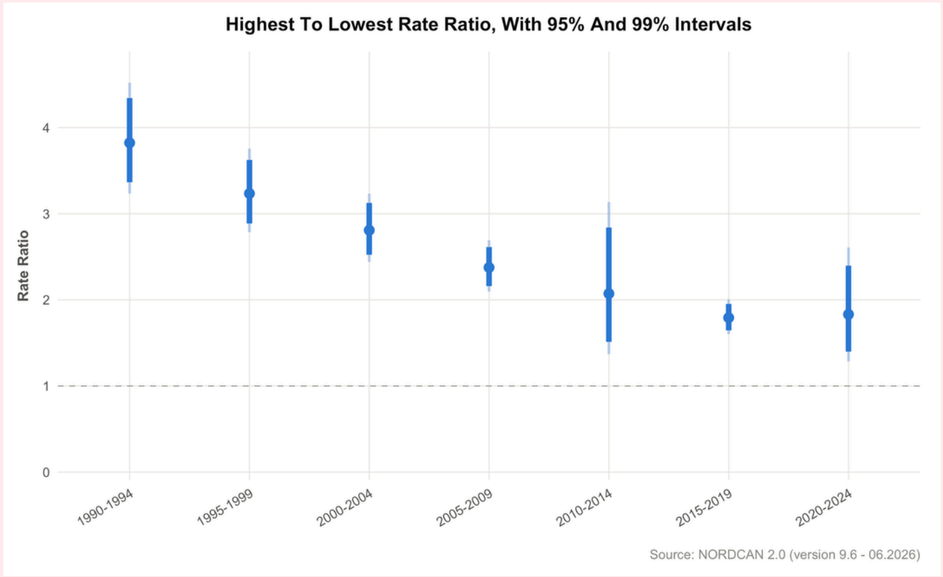
Only Fall
Iceland, -16%

Smallest Move
Denmark, +5%

Least Precise
Iceland, wide bars

The Gap Fell From 3.8 To 1.8

Highest To Lowest Rate Ratio, 95% And 99% Intervals



Period	Highest / Lowest	Ratio
1990-94	Denmark / Finland	3.82
1995-99	Denmark / Finland	3.23
2000-04	Norway / Finland	2.81
2005-09	Norway / Finland	2.38
2010-14	Norway / Iceland	2.07
2015-19	Denmark / Finland	1.79
2020-24	Denmark / Finland	1.83

1.79

Lowest point, in 2015-19

5 of 6

Steps at which the ratio fell

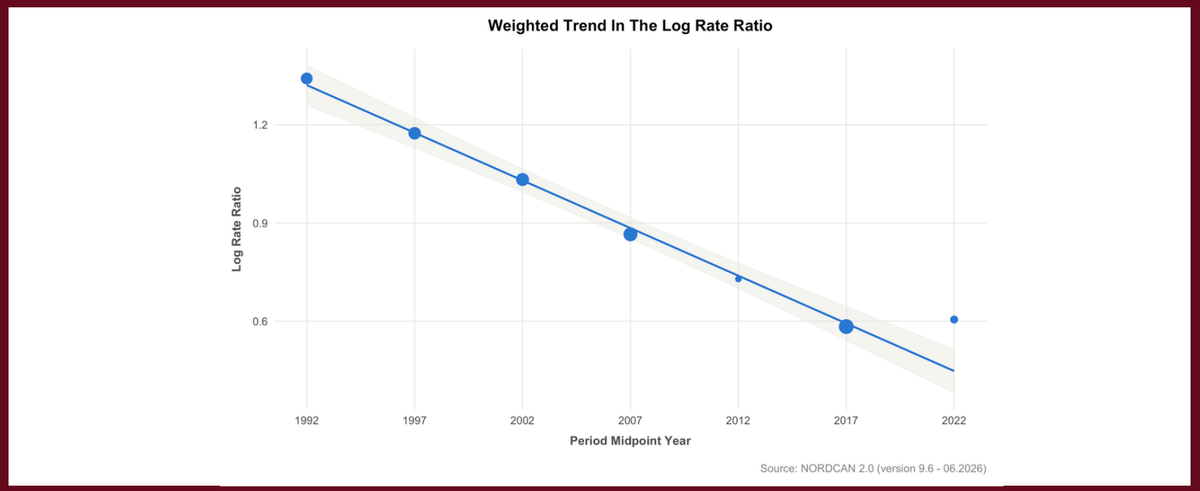
1.78

Raw ratio in 2020-24, from 3.99 in 1990-94

A Steady Decline, Not A Single Drop

The ratio falls in almost every period, and the first and last periods' intervals do not overlap.

Log Of The Rate Ratio, With Weighted Fitted Line



What The Fitted Line Says

Measure	Value
Average change in the ratio	-2.9% a year
95% interval	-3.2% to -2.5% a year
Fall over the 30 years between first and last midpoints	About 58%
Fitted ratio, 2020-24	About 1.56 (observed 1.83)

3.37 First period's lowest plausible ratio (95%)

VS

2.40 Last period's highest plausible ratio (95%)

What This Shows, And What It Does Not

From 1990 to 2024, the highest-to-lowest Nordic incidence ratio for testicular cancer fell from 3.82 to 1.83, and the narrowing is larger than the uncertainty around it.

Solid Results

1. The size of the narrowing: 3.82 to 1.83. The first and last intervals do not overlap at 95% or 99%.
2. It survives swapping every near-tied label: the slope moves from -0.0291 to -0.0293, the ratios by at most 0.035.
3. It survives standardisation: the raw ratio runs 3.99 to 1.78.

Less Solid Results

1. Who sits at each end in some periods: Denmark and Norway are 0.03 apart in 2015-19, Finland and Iceland about 0.1 apart in 2010-14 and 2020-24.
2. The ratio picks the highest and lowest country each period, and the intervals do not allow for that choice.
3. Seven periods is a short series, and Iceland, the least precise country, is an extreme in the two least precise periods.

Open Questions

Why the gap narrowed is a separate question. This project does not take it up.

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17th of October, 2026

Thank You.

Sources And Attribution

Association of the Nordic Cancer Registries / International Agency for Research on Cancer. NORDCAN 2.0, data version 9.6 (06.2026). https://nordcan.iarc.fr. Accessed 20 September 2026.

Tichanek et al. (2023). Cancers (Basel). DOI: 10.3390/cancers15174261.

Standard population: European Standard Population 2013, via the PHEindicatormethods R package.

Code and results: github.com/Rlvtick/nordcan-testicular-incidence